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Free Vibration of Composite Beams Including Rotary Inertia and Shear Deformation

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ABSTRACT

Exact solutions are presented for the free vibration of symmetrically laminated composite beams. First-order shear deformation and rotary inertia have been included in the analysis. The solution procedure is applicable to arbitrary boundary conditions. Results have been presented to demonstrate the effect of shear deformation, material anisotropy and boundary conditions on the natural frequencies of advanced composite beams.

INTRODUCTION

Renewed interest in laminated composite beams as robot arms, helicopter blades, and in numerous other applications warrants a better understanding of vibration characteristics of beams. During the last decade, several authors have tried to predict the natural frequencies of laminated plates,[1,2] but there have been few predictions for beams. Miller and Adams[3] studied the vibration characteristics of orthotropic clamped–free beams without including shear deformation and rotary inertia. A beam theory based on classical lamination theory has been presented by Vinson and Sierakowski.[4] An exact solution for the free vibration of simply supported composite beams without shear deformation and rotary inertia has also been reported in Ref. 4. The effects of transverse shear deformation are pronounced for composites, because of the high ratio of extensional modulus to the transverse shearing modulus, and classical lamination theory is inadequate for accurate prediction of natural

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frequencies. Teoh and Huang[5] have presented a theoretical analysis of the free vibrations of orthotropic clamped–free beams, using the energy approach including both shear and rotary inertia. Recently, Chen and Yang[6] presented a finite element analysis of symmetrically laminated composite beams to predict bending and free vibration, including the effect of shear deformation. In the present work, exact solutions have been presented for shear flexible symmetrically laminated composite beams. The solution method presented is applicable to arbitrary boundary conditions and the results can be used as a reference to approximate solutions.

MATHEMATICAL FORMULATION

A multilayered composite beam, as shown in Fig. 1, is considered. Following the procedure similar to that presented in Ref. 7 for laminated plates, the assumed displacement field for the composite beam based on first-order shear deformation theory takes the form

$$\bar{u}(x,z,t) = u(x,t) + z\psi(x,t) \qquad (1)$$

$$\bar{w}(x,z,t) = w(x,t)$$

where u and w are the displacements of a point on the midplane and ψ is the rotation of the normal to the midplane.

The strain displacement equations are given by

$$\varepsilon_x = \varepsilon_x^0 + z\kappa_x$$

$$\gamma_{xz} = \psi + \frac{\partial w}{\partial x} \qquad (2)$$

where $\varepsilon_x^0 = \partial u/\partial x$ and $\kappa_x = \partial\psi/\partial x$.

The constitutive equation can be written as

$$\begin{Bmatrix} N_x \\ M_x \end{Bmatrix} = \begin{bmatrix} A_{11} & B_{11} \\ B_{11} & D_{11} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \kappa_x \end{Bmatrix} \qquad (3)$$

$$Q_{xz} = A_{55}\gamma_{xz} \qquad (4)$$

where

$$(N_x, M_x) = \int_A \sigma_x(1, z)\,\mathrm{d}A \qquad (5)$$

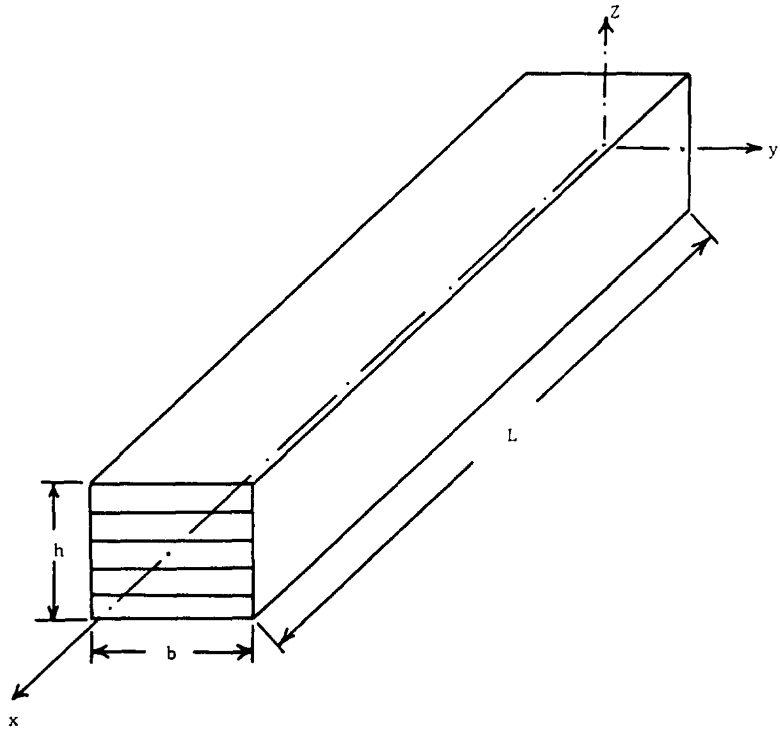

Fig. 1. Geometry of a composite beam.

Here A_{11}, B_{11} and D_{11} denote the extensional, flexural–extensional coupling and flexural stiffnesses of the laminated beam, and are given by

$$(A_{11}, B_{11}, D_{11}) = b \int_{-h/2}^{h/2} \bar{Q}_{11}(1, z, z^2)\,\mathrm{d}z \tag{6}$$

and A_{55} is the shear coefficient, defined by

$$A_{55} = bK \int_{-h/2}^{h/2} \bar{Q}_{55}\,\mathrm{d}z \tag{7}$$

where $\bar{Q}_{11}$ and $\bar{Q}_{55}$ are the transformed material constants given by[8]

$$\bar{Q}_{11} = Q_{11}\cos^4\theta + Q_{22}\sin^4\theta + 2(Q_{12} + 2Q_{66})\sin^2\theta\cos^2\theta$$
$$\bar{Q}_{55} = G_{13}\cos^2\theta + G_{23}\sin^2\theta$$

and K is the shear correction factor.

EQUATIONS OF MOTION

The dynamic version of the principle of virtual work for the present problem is

$$\delta\pi = 0 = \int_0^t \int_0^L \{N_x \delta\varepsilon_x^0 + M_x \delta\kappa_x + Q_{xz}\delta\gamma_{xz} + (I_1\dot{u} + I_2\dot{\psi})\delta\dot{u} + I_1\dot{w}\delta\dot{w} + (I_3\dot{\psi} + I_2\dot{u})\delta\dot{\psi}\}\,\mathrm{d}x\,\mathrm{d}t \tag{8}$$

where

$$(I_1, I_2, I_3) = b\int_{-h/2}^{h/2} \rho(1, z, z^2)\,\mathrm{d}z$$

The equations of motion for the laminated composite beam can be obtained by substituting eqns (1), (3) and (4) in eqn (8), integrating the displacement gradients by parts and setting the coefficients of δu, δw and $\delta\psi$ to zero separately. For a laminated beam with midplane symmetry, B_{11} is equal to zero and the in-plane displacement can be assumed to be negligible compared with the flexure-induced displacement. The equations of motion reduce to

$$A_{55}\left(\frac{\partial\psi}{\partial x} + \frac{\partial^2 w}{\partial x^2}\right) - I_1\frac{\partial^2 w}{\partial t^2} = 0 \tag{9}$$

$$D_{11}\frac{\partial^2\psi}{\partial x^2} - A_{55}\left(\psi + \frac{\partial w}{\partial x}\right) - I_3\frac{\partial^2\psi}{\partial t^2} = 0 \tag{10}$$

Equations (9) and (10) are used to determine the free vibration of the composite beam.

SOLUTION PROCEDURE

The harmonic solution can be assumed in the form

$$w = W\mathrm{e}^{i\omega t} \tag{11}$$

$$\psi = \Psi\mathrm{e}^{i\omega t} \tag{12}$$

where ω is the circular frequency.

Using eqns (11) and (12), eqns (9) and (10) can be expressed as

$$\frac{d^2 W}{d\xi^2} + a^2 c^2 W + L \frac{d\Psi}{d\xi} = 0 \tag{13}$$

$$c^2 \frac{d^2 \Psi}{d\xi^2} - (1 - a^2 b^2 c^2)\Psi - \frac{1}{L}\frac{dW}{d\xi} = 0 \tag{14}$$

where

$$a^2 = \frac{I_1 L^4 \omega^2}{D_{11}}$$

$$b^2 = \frac{I_3}{I_1 L^2}$$

$$c^2 = \frac{D_{11}}{A_{55} L^2}$$

and $\xi = x/L$.

Equations (13) and (14) can be transformed into two uncoupled differential equations by eliminating W or Ψ as follows:

$$\frac{d^4 W}{d\xi^4} + a^2(b^2 + c^2)\frac{d^2 W}{d\xi^2} - a^2(1 - a^2 b^2 c^2) W = 0 \tag{15}$$

$$\frac{d^4 \Psi}{d\xi^4} + a^2(b^2 + c^2)\frac{d^2 \Psi}{d\xi^2} - a^2(1 - a^2 b^2 c^2)\Psi = 0 \tag{16}$$

Solutions to eqns (15) and (16) can be expressed as[9]

Case (i) If $[(b^2 - c^2)^2 + 4/a^2]^{1/2} > (b^2 + c^2)$

$$W = A_1 \cosh a\alpha\xi + A_2 \sinh a\alpha\xi + A_3 \cos a\beta\xi + A_4 \sin a\beta\xi \tag{17}$$

$$\Psi = B_1 \sinh a\alpha\xi + B_2 \cosh a\alpha\xi + B_3 \sin a\beta\xi + B_4 \cos a\beta\xi \tag{18}$$

where

$$\begin{matrix}\alpha \\ \beta\end{matrix} = \frac{1}{\sqrt{2}}\left\{ \begin{matrix}- \\ +\end{matrix}(b^2 + c^2) + [(b^2 - c^2)^2 + 4/a^2]^{1/2} \right\}^{1/2}$$

Case (ii) If $[(b^2-c^2)^2+4/a^2]^{1/2}<(b^2+c^2)$

$$W = A_1\cos a\alpha'\xi + iB_2\sin a\alpha'\xi + A_3\cos a\beta\xi + A_4\sin a\beta\xi \tag{19}$$

$$\Psi = iB_1\sin a\alpha'\xi + B_2\cos a\alpha'\xi + B_3\sin a\beta\xi + B_4\cos a\beta\xi \tag{20}$$

where $\alpha' = \alpha/i$.

The values of the constants are related by the coupled eqns (13) and (14):

$$\begin{aligned}
B_1 &= -\frac{a}{L}\frac{(\alpha^2+c^2)}{\alpha}A_1\\
B_2 &= -\frac{a}{\mathrm{L}}\frac{(\alpha^2+c^2)}{\alpha}A_2\\
B_3 &= \frac{a}{L}\frac{(\beta^2-c^2)}{\alpha}A_3\\
B_4 &= -\frac{a}{L}\frac{(\beta^2-c^2)}{\beta}A_4
\end{aligned} \tag{21}$$

The six commonly encountered boundary conditions of the beam considered are:

Simply supported (SS)

$$\text{At } \xi = 0,1 \qquad W = 0 \qquad \frac{\mathrm{d}\Psi}{\mathrm{d}\xi} = 0 \tag{22}$$

Clamped–clamped (CC)

$$\text{At } \xi = 0,1 \qquad W = 0 \qquad \Psi = 0 \tag{23}$$

Free–free (FF)

$$\text{At } \xi = 0,1 \qquad \frac{\mathrm{d}\Psi}{\mathrm{d}\xi} = 0 \qquad \frac{1}{L}\frac{\mathrm{d}W}{\mathrm{d}\xi} + \Psi = 0 \tag{24}$$

The other three boundary conditions are the combinations of the above boundary conditions. They are supported–free (SF), clamped–free (CF) and clamped–supported (CS).

Using case (i) and case (ii) solutions for the six types of beams considered leads to the following matrix equation:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ L_1 & L_2 & L_3 & L_4 \\ L_5 & L_6 & L_7 & L_8 \\ L_9 & L_{10} & L_{11} & L_{12} \end{bmatrix} \begin{Bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \tag{25}$$

For brevity, the coefficients L_1, L_2, etc., for only SS and CF beams corresponding to case (i) solution are given in the Appendix. The frequency equation for the six types of beams corresponding to case (i) are

$$\text{SS} \quad \sin a\beta = 0 \tag{26}$$

$$\text{CC} \quad 2 - 2\cosh a\alpha \cos a\beta + \frac{a}{(1 - a^2b^2c^2)^{1/2}} [a^2c^2(b^2 - c^2)^2 + (3c^2 - b^2)] \sinh a\alpha \sin a\beta = 0 \tag{27}$$

$$\text{FF} \quad 2 - 2\cosh a\alpha \cos a\beta + \frac{a}{(1 - a^2b^2c^2)^{1/2}} [a^2c^2(b^2 - c^2)^2 + (3b^2 - c^2)] \sinh a\alpha \sin a\beta = 0 \tag{28}$$

$$\text{SF} \quad \frac{\alpha}{\beta} \tanh a\alpha - \frac{\alpha^2 + b^2}{\alpha^2 + c^2} \tan a\beta = 0 \tag{29}$$

$$\text{CF} \quad 2 + [a^2(b^2 - c^2) + 2] \cosh a\alpha \cos a\beta - \frac{a(b^2 + c^2)}{(1 - a^2b^2c^2)^{1/2}} \sinh a\alpha \sin a\beta = 0 \tag{30}$$

$$\text{CS} \quad \frac{\alpha(\alpha^2 + b^2)}{\beta(\alpha^2 + c^2)} \tanh a\alpha - \tan a\beta = 0 \tag{31}$$

Frequency equations corresponding to case (ii) are

$$\text{SS} \quad \sin a\beta = 0 \tag{32}$$

$$\text{CC} \quad 2 - 2\cos a\alpha' \cos a\beta + \frac{a}{(a^2b^2c^2 - 1)^{1/2}} [a^2c^2(b^2 - c^2)^2 + (3c^2 - b^2)] \sin a\alpha' \sin a\beta = 0 \tag{33}$$

$$\text{FF}\qquad 2 - 2\cos a\alpha'\cos a\beta + \frac{a}{(a^2b^2c^2-1)^{1/2}}$$

$$[a^2b^2(b^2-c^2)^2 + (3b^2-c^2)]\sin a\alpha'\sin a\beta = 0 \tag{34}$$

$$\text{SF}\qquad \frac{\alpha'}{\beta}\tan a\alpha' + \frac{\alpha^2+b^2}{\alpha^2+c^2}\tan a\beta = 0 \tag{35}$$

$$\text{CF}\qquad 2 + [a^2(b^2-c^2)^2+2]\cos a\alpha'\cos a\beta$$

$$-\frac{a(b^2+c^2)}{(a^2b^2c^2-1)^{1/2}}\sin a\alpha'\sin a\beta = 0 \tag{36}$$

$$\text{CS}\qquad \frac{\alpha'(\alpha^2+b^2)}{\beta(\alpha^2+c^2)}\tan a\alpha' + \tan a\beta = 0 \tag{37}$$

It should be noted that the frequency equations corresponding to case (i) are applicable when ($\omega^2 < A_{55}/I_3$) and those corresponding to case (ii) are applicable when ($\omega^2 > A_{55}/I_3$).

NUMERICAL RESULTS

The shear correction factor K is taken as 5/6 to account for the parabolic variation of transverse shear stresses. The following AS/3501-6 graphite–epoxy material properties are used in the example problems considered:

$$E_1 = 21{\cdot}0\times10^6\,\text{psi},\quad E_2 = 1{\cdot}4\times10^6\,\text{psi},\quad G_{23} = 0{\cdot}5\times10^6\,\text{psi}$$

$$G_{12} = G_{13} = 0{\cdot}6\times10^6\,\text{psi},\quad \nu_{12} = 0{\cdot}3,\quad \rho = 0{\cdot}13\times10^{-3}\,\text{lb s}^2\,\text{in}^{-4}$$

To justify the validity of the present formulation, Table 1 shows the comparison of the first five natural frequencies of long–thin (L/h = 120) and short–thick (L/h = 15) simply supported orthotropic (0°) beams. Beam width is taken as unity for all the problems considered. For long–thin beams, the shear deformation theory (SDT) and classical lamination theory (CLT) give almost the same result, indicating no effect of shear deformation. For short–thick beams, the CLT solution over-predicts the natural frequencies and the shear effect is pronounced for higher modes. In Table 2, the first five non-dimensional frequencies of four layer symmetric cross-ply beams with different boundary conditions are presented. Table 3 shows the non-dimensional fundamental frequencies of four-layer symmetric angle-ply beams. It is clear from Table 3 that the frequencies decrease with increase in fiber orientation for all the boundary conditions of the beam.

TABLE 1
Comparison of Natural Frequencies of a Simply Supported Orthotropic (0°) Graphite–Epoxy Beam

L/h	Mode	$f(kHz)$ SDT Present	$f(kHz)$ CLT[4]
120	1	0·051	0·051
(h = 0·25)	2	0·203	0·203
	3	0·457	0·457
	4	0·812	0·812
	5	1·269	1·269
15	1	0·755	0·813
(h = 1)	2	2·548	3·250
	3	4·716	7·314
	4	6·960	13·002
	5	9·194	20·316

TABLE 2
Non-dimensional Frequencies [$\bar{\omega} = \omega L^2 \sqrt{(\rho/E_1 h^2)}$] of [0/90/90/0] Cross-Ply Beams (L/h = 15)

Beam type \ Mode no.	1	2	3	4	5
SS	2·5023	8·4812	15·7558	23·3089	30·8386
CC	4·5940	10·2906	16·9659	24·0410	31·2874
FF	3·8576	10·5367	18·1546	25·8874	33·5243
SF	5·5540	12·7657	20·6905	28·5560	36·2835
CF	0·9241	4·8925	11·4400	18·6972	26·2118
CS	3·5254	9·4423	16·3839	23·6850	31·0659

TABLE 3
Non-dimensional Fundamental Frequencies [$\bar{\omega} = \omega L^2 \sqrt{(\rho/E_1 h^2)}$] of [$\theta/-\theta/-\theta/\theta$] Angle-Ply Beams ($L/h$ = 15)

Beam type \ θ (deg.)	0	15	30	45	60	75	90
SS	2·6560	2·5105	2·1032	1·5368	1·0124	0·7611	0·7320
CC	4·8487	4·6635	4·0981	3·1843	2·1984	1·6815	1·6200
FF	4·0931	3·8728	3·2530	2·3841	1·5738	1·1840	1·1389
SF	5·8923	5·5774	4·6894	3·4407	2·2730	1·7105	1·6453
CF	0·9820	0·9249	0·7678	0·5551	0·3631	0·2723	0·2619
CS	3·7305	3·5593	3·0573	2·3032	1·5511	1·1753	1·1312

CONCLUSIONS

The natural frequencies of symmetrically laminated composite beams have been predicted using first-order shear deformation theory. Numerical results have been presented for beams with various commonly encountered boundary conditions and can be used as a reference to approximate solutions.

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APPENDIX

Simply supported beam

$$L_1 = -\frac{a^2}{L}(\alpha^2 + c^2) \qquad L_2 = 0$$

$$L_3 = \frac{a^2}{L}(\beta^2 - c^2) \qquad L_4 = 0$$

$$L_5 = \cosh a\alpha \qquad L_6 = \sinh a\alpha$$

$$L_7 = \cos a\beta \qquad L_8 = \sin a\beta$$

$$L_9 = -\frac{a^2}{L}(\alpha^2 + c^2)\cosh a\alpha \qquad L_{10} = -\frac{a^2}{L}(\alpha^2 + c^2)\sinh a\alpha$$

$$L_{11} = \frac{a^2}{L}(\beta^2 - c^2)\cos a\beta \qquad L_{12} = \frac{a^2}{L}(\beta^2 - c^2)\sin a\beta$$

Clamped–free beam

$$L_1 = 0 \qquad L_2 = -\frac{\alpha^2 + c^2}{\alpha}$$

$$L_3 = 0 \qquad L_4 = -\frac{\beta^2 - c^2}{\beta}$$

$$L_5 = -(\alpha^2 + c^2)\cosh a\alpha \qquad L_6 = -(\alpha^2 + c^2)\sinh a\alpha$$

$$L_7 = (\beta^2 - c^2)\cos a\beta \qquad L_8 = (\beta^2 - c^2)\sin a\beta$$

$$L_9 = \left(\alpha - \frac{\alpha^2 + c^2}{\alpha}\right)\sinh a\alpha \qquad L_{10} = \left(\alpha - \frac{\alpha^2 + c^2}{\alpha}\right)\cosh a\alpha$$

$$L_{11} = -\left(\beta - \frac{\beta^2 - c^2}{\beta}\right)\sin a\beta \qquad L_{12} = \left(\beta - \frac{\beta^2 - c^2}{\beta}\right)\cos a\beta$$